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HY USMLE REVIEW
PART III

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HY USMLE Review – Part III

- 72F + radical mastectomy 25 years ago + hard, raised purple lesions above the elbow; Dx? → lymphangiosarcoma (Stewart-Treves syndrome) → you don't have to agree that it's HY, but it's asked on the NBME → caused by chronic lymphatic insufficiency classically years after radical mastectomy.
- Neonate + spongy 1-cm red lesion on the chest; Dx? → strawberry hemangioma
- Strawberry hemangioma Tx? → don't treat; will grow slightly then regress spontaneously over a few years
- Neonate + large vascular lesion on the leg + thrombocytopenia; Dx? → Kasabach-Merritt syndrome (aka hemangioma with thrombocytopenia) → this is on the pediatric 2CK forms **three times** asked in different ways; students always say wtf and I have to explain that, yes, it's weird, but it's HY for some magical reason; this is not a strawberry hemangioma and requires surgical Tx.
- Neonate + large vascular lesion on the leg + thrombocytopenia; what is the cause of the thrombocytopenia? → answer = "platelet sequestration." I've memorized this from the NBMEs → similar to splenomegaly, which can cause thrombocytopenia from sequestration within the red pulp, the implication that the large vascular lesion of KMS is that platelets simply get caught within it.
- "Cherry red blood/lips" → CO poisoning
- "Brown blood" or "chocolate blood" → methemoglobinemia
- Kid with brown blood and they ask you the mechanism (answers are "upregulation of anti-proteinase 2" or "deficiency of cytochrome reductase B5") → answer = deficiency of cytochrome reductase B5; this is on the USMLE. I'm not fucking with you. And if you Wiki it, you'll see clear as day that they talk about congenital methemoglobinemia due to deficiency of cytochrome reductase B5.
- 22M + violaceous papules in a temporal distrubtion → answer = Sturge-Weber syndrome; student says "wtf? I thought that was associated with Port wine stain birthmark." Yeah, if we take a trip back to kindergarten, but you need to know it can present as **cutaneous papules** in a trigeminal nerve distribution.
- 44M alcoholic + winter + they show a pic of his feet and they're red; what electrolyte are we most worried about upon rewarming them → answer = hyperkalemia → alcoholics are notably susceptible to rhabo (ultra HY on the USMLE) → rhabdo causes myoglobin release, which is nephrotoxic and can cause acute tubular necrosis (potassium goes up); even if the patient doesn't get full-blown rhabdo

with ATN, reperfusion injury can cause O2 radical-mediated damage that induces cell lysis (increases K levels).

- 14M + ataxia + cognitive decline over a few months → answer = **glue, not** alcohol → no way a kid that young would get alcoholic cerebellar ataxia; this is on the NBME even if you find it stupid/weird.
- 16M found unconscious on floor in school bathroom + normal vitals + no eye findings + a little sluggish → answer = **butane (inhalant) toxicity** → caused by "dusters" / inhaling computer cleaner; **this is on the USMLE!!!**
- Medial malleolus ulcer + hyperpigmentation of lower legs; Dx? → chronic venous insufficiency
- Punched-out ulcer on foot + intermittent claudication; Dx? → arterial insufficiency (peripheral vascular disease)
- What causes venous insufficiency? → valvular incompetence (most commonly familial), resulting in venous reflux + insufficiency.
- What causes arterial insufficiency → atherosclerosis **(diabetes, followed by smoking, are the two most acceleratory risk factors; hypertension is the most common risk factor)**
- How do you Dx venous insufficiency? → **duplex ultrasound** of the calves showing stasis and/or occlusive disease (the latter may result from venous insufficiency or cause it)
- How do you Dx arterial insufficiency? → USMLE always wants **ankle-brachial indices (ABI) first** → after this is done, the answer **is Doppler ultrasound** of the calves (duplex ultrasound is the answer for venous) **or arteriography**; both of these latter answers are correct; they will not give you both; it will be one or the other.
- Tx for venous insufficiency → compression stockings
- Tx for varicose veins → compression stockings
- Varicose veins and venous insufficiency same thing? → varicose veins are one of the mere presentations of venous insufficiency, so yes, patients with varicose veins have venous insufficiency.
- 47F has varicose veins + painful palpable cord by the ankle (is the treatment compression stockings or subcutaneous enoxaparin; both are listed) → answer = subcutaneous enoxaparin because this is superficial thrombophlebitis.
- Tx for arterial insufficiency → exercise regimen first, THEN cilostazol (phosphodiesterase 3 inhibitor)

- What must you do before starting the exercise regimen in the Tx of arterial insufficiency → **ECG stress test** to ascertain patient's exercise tolerance.
- What is patient has abnormal baseline ECG (e.g., BBB) → do echo stress test instead.
- What if the patient can't exercise → do dobutamine-echo stress test
- What if the patient gets stable angina after merely walking up a flight of stairs → skip stress test and go straight to myocardial perfusion scan (myocardial scintigraphic assay); this is answer on the NBME.
- Patient has severe ischemia on stress test or myocardial perfusion scan → do coronary angiography → then do coronary artery bypass grafting if three-vessel disease, OR two-vessel disease + diabetic, OR single-vessel disease if it's the left main coronary.
- Patient with CVD is on various medications + has hyperkalemia; why? → ACEi, ARB, and spironolactone all can cause hyperkalemia.
- Patient with CVD is on various medications + hypokalemia; why? → furosemide (Loop diuretic)
- When do we start patients on furosemide? → to fluid unload (dyspnea in heart failure or peripheral edema)
- Patient is started on furosemide + still has fluid overload; what's the next diuretic to use → spironolactone (this is really HY on the USMLE and is on Steps 1 and 2CK NBMEs) → essentially furosemide causes increased K wasting, so we must give a potassium-sparing diuretic to balance the effect (spironolactone).
- What's the MOA of spironolactone → aldosterone receptor antagonist.
- Side-effects of spironolactone → hyperkalemia; gynecomastia.
- When do we give patients spironolactone apart from as a step-up from Loops? → added onto heart failure management after a patient is already on ACEi (or ARB) + beta-blocker. In other words, for heart failure: give ACEi (or ARB) first, then add beta-blocker, then add spironolactone.
- Major side-effects of beta-blockers → depression + sexual dysfunction (avoid in these patients)
- Major side-effect of naproxen → fluid retention (edema) due to increased renal retention of sodium.
- What is naproxen? → NSAID that the USMLE is obsessed with for some reason.
- Why might NSAIDs cause fluid retention / renal retention of sodium? → knocking out COX → decreased prostaglandin synthesis → decreased renal afferent arteriolar dilatation → decreased

renal blood flow → PCT of kidney compensates for perceived low blood volume by increasing Na reabsorption → water follows sodium → edema.

- BUN/Cr ratio in someone with NSAID-induced interstitial nephropathy → >20 (pre-renal).
- What kind of vignette is a patient classically on naproxen? → osteoarthritis (OA).
- How do we treat OA? → weight loss (biggest risk factor is obesity), then acetaminophen, then NSAIDs → sometimes USMLE will mention naproxen being taken by OA patient to illustrate this common scenario of poor self-management → acetaminophen recommended before NSAIDs because the renal damage caused by NSAID use.
- What are the four beta-blockers that decrease mortality in heart failure? → Metoprolol XR (extended release) + carvedilol + nebivolol + bisoprolol → USMLE will never ask “extended-release”; they’ll just want metoprolol, but cardiologists will spasm out if you say regular metoprolol without specifying extended release; the idea being: we don’t give drugs like propranolol, atenolol, etc., for heart failure because they don’t improve mortality.
- When do we use “regular” metoprolol → classically first-line for atrial fibrillation rate control.
- When do we use atenolol → stage-fright
- When do we use propranolol → migraine prophylaxis, tachycardia in hyperthyroidism (decreases peripheral conversion of T4 to T3), esophageal varices prophylaxis, akathisia caused by anti-psychotics, hypertrophic cardiomyopathy to increase preload (HOCM and MVP are the two murmurs that get worse with low preload), essential tremor (AD familial tremor; patients self-medicate with alcohol, which decreases tremor; but propranolol can also be used for other tremors; social phobia (different from stage-fright).
- Patient takes medication for muscle pain relief + gets wheezing (which should be avoided, acetaminophen or aspirin; both are listed) → answer = aspirin (aspirin-induced asthma)
- 27F + intermittent headaches + blurry vision; Dx? → optic neuritis (multiple sclerosis) → student says “why the headaches?” Yeah, I know. Weird. But it’s on the NBME. You need to know optic neuritis is HY in MS and means inflammation of cranial nerve II → presents as blurry vision, or change in color vision, or Marcus Gunn pupil (relative afferent pupillary defect)
- Most specific eye finding in MS → medial longitudinal fasciculus (MLF) syndrome → aka internuclear ophthalmoplegia (INO) → when you abduct to one side, you activate CN VI on that side, which

requires the contralateral CN III to activate in order to adduct → the side that cannot adduct is the side that's fucked up; the normal side will have nystagmus.

- Tx for acute flare of MS → IV steroids (oral is wrong and can make flares worse).
- Tx between flares of MS (the patient must by asymptomatic) → IFN-beta (interferon beta).
- Tx for spasticity in MS → baclofen.
- MOA of baclofen → GABA-B receptor agonist (USMLE loves baclofen).
- Dx of MS → choose MRI over CSF IgG oligoclonal bands if both are listed.
- Urinary incontinence in MS → urge incontinence (hyperactive detrusor / detrusor instability)
- 65F + unilateral temporal headache + blurry vision; Dx? → temporal arteritis
- Tx for TA → immediate IV methylprednisolone (IV steroids) to prevent blindness, **then** do temporal artery biopsy; **do not** choose biopsy before steroids.
- What other condition is TA associated with? → polymyalgia rheumatica (in fact, they're considered to be on the same disease spectrum, rather than as two inherently distinct conditions)
- 65F + unilateral temporal headache + muscle pain/stiffness → temporal arteritis + polymyalgia rheumatica
- Tx for PR if patient doesn't also have temporal arteritis → oral steroids are okay bc not an emergency.
- Polymyalgia rheumatica vs polymyositis → PR presents with stiffness/pain **generally without weakness;** polymyositis can present with pain, **but it also has proximal muscle weakness.** Both conditions can present with elevations of ESR and CRP, so these aren't reliable for Dx.
- Dx of PR → clinical; there's no specific test.
- Dx of polymyositis → same as with dermatomyositis, do anti-Jo1 and -Mi2 antibodies, then do confirmatory muscle biopsy.
- Oral + genital ulcers → Behcet syndrome
- Violaceous rash around the eyelids → heliotrope rash → Dx = dermatomyositis (don't confuse that with malar rash of SLE).
- Violaceous papules on knuckles + cracked/dry palms = dermatomyositis (Gottron papules + mechanics' hands).
- Rash around back of neck + top of back + adult with proximal muscle weakness (difficulty standing from chair unassisted) → shawl rash → dermatomyositis.

- Dx of dermatomyositis → anti-Jo1 or -Mi2 antibodies first → muscle biopsy confirmatory.
- Polymyositis = dermatomyositis but without the skin findings.
- USMLE wants you to know that in dermatomyositis, there's specifically an increased risk of what? → malignancy (non-Hodgkin lymphoma) → that being said, autoimmune diseases in general increase the risk of NHL → so if you have, e.g., patient with SLE with irregular ring-enhancing lesion on head CT, answer = primary CNS lymphoma, not toxoplasmosis.
- Tx for flare of polymyositis/dermatomyositis → oral corticosteroids.
- Antibodies in pemphigus vulgaris? → anti-desmoglein (anti-desmosome) → desmosomes enable adjacent cell-to-cell adhesion.
- Antibodies in bullous pemphigoid → anti-hemidesmosome → hemidesmosomes secure the dermis to the epidermal basal layer.
- Which one is worse, bullous pemphigoid or pemphigus vulgaris? → pemphigus vulgaris; why? → PV gets bullae on the skin that rupture with friction (Nikolsky sign) + scar → oral mucosal involvement is more prevalent in PV.
- What kind of immunofluorescence on biopsy in PV vs BV? → Linear in BV; net-like in PV.
- What else is associated with linear immunofluorescence on the USMLE? → Goodpasture syndrome (on biopsy of the kidney or alveoli).
- Tx for PV and BV → oral corticosteroids
- Mechanism for Goodpasture syndrome? → antibodies against type IV collagen → "2, 3, 4… 2, 3, 4… 2, 3, 4. The Goodpasture is marching in the field, 2, 3, 4!"→ Type 2 hypersensitivity against the alpha-3 chains of type 4 collagen.
- Hematuria + hemoptysis + "head-itis" (mastoiditis, sinusitis, otitis, nasal septal perforation) → Wegener granulomatosis
- Annoying new name for Wegener → granulomatosis with polyangiitis
- Dx of Wegener → c-ANCA (anti-PR3; anti-proteinase 3)
- Asthma + eosinophilia → Churg-Strauss
- Annoying new name for CS → eosinophilic granulomatosis with polyangiitis
- Dx of CS → p-ANCA (anti-MPO; anti-myeloperoxidase)

- Hematuria in isolation + p-ANCA in serum → microscopic polyangiitis (MP)
- Severe renal disease in Wegener or Goodpasture or MP → rapidly progressive glomerulonephritis (crescentic)
- 44M + hematuria + hemoptysis → Goodpasture syndrome
- 44M + hematuria + hemopytisis + head-itis → Wegener
- What is polyarteritis nodosa → medium-vessel vasculitis characterized by immune complex deposition in vascular walls and fibrinoid necrosis
- Polyarteritis nodosa is associated with what infection? → 30% of patients are HepB positive
- What do you see on renal artery angiogram in PN → "beads on a string" (similar to fibromuscular dysplasia, although completely unrelated diseases).
- Which vessels are notably not affected in PN → pulmonary vessels (USMLE likes this detail).
- Tx for PN → oral corticosteroids
- 32F + arthritis + mouth ulcer + low platelets; Dx? → SLE
- Most common presentation finding in SLE → arthritis (>90%)
- Woman 20s-40s + arthritis + thrombocytopenia → SLE
- Woman 20s-40s + arthritis + mouth ulcer + circular skin lesions → SLE
- Malar rash + low RBCs + low WBCs + low platelets; mechanism for low cell lines? → increased peripheral destruction (antibodies against hematologic cells lines seen in SLE; isolated thrombocytopenia most common)
- Tx of SLE flare → steroids
- SLE + red urine; Dx? → lupus nephritis, more specifically, diffuse proliferative glomerulonephritis (DPGN)
- Histology of DPGN → wire looping capillary pattern
- Tx of lupus nephritis → mycophenolate mofetil
- Tx of discoid lupus → hydroxychloroquine
- Most specific Abs for SLE → anti-Smith (RNP), not anti-dsDNA
- Which Abs go up in acute SLE flares → anti-dsDNA (and C3 goes down)

- Drug-induced lupus Abs → anti-histone
- Drugs that cause DIL → Mom is HIPP → Minocycline, Hydralazine, INH, Procainamide, Penicillamine
- Viral infection + all three cell-lines are down → viral-induced aplastic anemia
- Viral-induced aplastic anemia; next best step in Dx? → bone marrow aspiration
- Viral-induced aplastic anemia; mechanism? → defective bone marrow production (contrast with SLE)
- 80F has catheter; how to best decrease infection risk in this patient (all answers listed sound reasonable) → correct answer = hand washing
- 17M has mononucleosis; how to best decrease risk of transmission? → answer = hand washing ("huh, I thought it was just kissing + sharing cups n stuff." I agree with you. But the USMLE wants handwashing.
- 72M + intermittent claudication + absent distal pulses + Hx of coronary artery bypass grafting + high BP that's been gradually increasing past two years; Dx? → renal artery stenosis
- 32F + high BP + high aldosterone/renin → fibromuscular dysplasia (tunica media proliferation in renal arteries) → this is **not** renal artery stenosis → if you say "renal artery stenosis," that means atherosclerosis.
- Increased creatinine following medication administered to someone with renal artery stenosis; what was the drug? → ACEi or ARB
- Tx for RAS + FMD → initially medical therapy with cautious use of ACEi or ARB; definitive is renal angioplasty + stenting; FMD is not curable.
- Dx of malaria → thick + thin blood smears, **not** antibody titer for Plasmodium species
- Girl goes to Africa + is taking chloroquine prophylaxis → gets malaria anyway; why? Is the answer non-adherence or resistance to chloroquine? → answer = resistance to chloroquine
- What is the MOA of chloroquine? → inhibits Plasmodium heme polymerase.
- Which malaria type is the worst and why? → P. falciparum because it causes cerebral malaria.
- Fever cycles and malaria? → P. vivax/ovale have fever every 48 hours; P. malariae every 72 hours; P. falciparum causes sporadic/unpredictable fever.
- Which drug is often given to people for malaria prophylaxis who go to chloroquine-resistant areas? → mefloquine

- Girl goes to Africa + gets malaria + she is then treated after the fact with atovoquine + proguanil and the presentation subsides; then one month later she has malaria again; why? → answer = reactivation of non-erythrocytic form of organism” → she has P. vivax or P. ovale → cause hypnozoites which are an intra-hepatic form of the disease.
- Question literally asks you point-blank why a patient is given primaquine → answer = “primaquine kills hypnozoites” or “kills extra-erythrocytic forms.”
- Why does sickle cell confer resistance to malaria → decreased RBC lifespan (malaria needs normal RBC lifespan to complete life cycle)
- Travel + bloody diarrhea + RUQ pain; Dx? → liver abscess due to Entamoeba histolytica
- How do you Tx the liver abscess? → answer = percutaneous drainage **BEFORE** antibiotics
- Abx for E. histolytica? → metronidazole + iodoquinol (latter kills intraluminal parasite); paromomycin may also be given.
- Watery diarrhea in immunocompromised patient → Cryptosporidium parvum
- How is Giardia transmitted (is the answer “water-borne” or “fecal-oral”?); answer = water-borne
- Steatorrhea in guy who went swimming or scuba diving → Giardia
- Tx for Giardia → metronidazole
- Fever + periorbital edema + muscle aches + went to a BBQ; Dx? → Trichinella spiralis → this is a classic triad seen in trichinosis.
- How do you get trichinosis? → bear meat (yes, Alaska is still in the United States and people hunt polar bear) or pork (pork nematode, not cestode)
- What is nematode vs cestode vs trematode? → nematode is roundworm; cestode tapeworm; trematodes are flukes.
- Nematode from pork → Trichinella spiralis
- Cestode (tapeworm) from pork → Taenia solium
- What does T. solium cause? → cysticersosis (muscle cysts) or neurocysticercosis (brain cysts)
- “Swiss cheese appearance” of brain in someone who traveled abroad → neurocysticercosis
- Single cystic lesion seen on brain CT in someone who went to Mexico → neurocysticercosis

- Tx for cysticersosis / neurocysticercosis → praziquantel or albendazole (the USMLE will never give you both and make you choose between them; for anti-helminth drugs questions, the correct answer will be the only anti-helminth drug listed).
- Which helminth causes visceral larva migrans? → Toxicara canis
- Worm in the eye? → Loa Loa
- How do you get Loa Loa → deer, horse, or mango fly
- HIV patient with lobar pneumonia (is the answer PJP or S. pnuemo?) → answer = Strep pneumo → PJP presents as bilateral interstitial infiltrates + groundglass appearance on CXR; S. pneumo is lobar → sort of a trick Q similar to asbestos (i.e., bronchogenic carcinoma still more likely than mesothelioma in pt with Hx of asbestos exposure; well S. pneumo still more common than PJP in immunocompromised pts). The key though is the lobar vs bilateral presentation as mentioned above.
- Upper lobe nodular density in immunocompromised patient → Aspergilloma
- Dx of Aspergilloma → open lung biopsy
- 13F with irregular periods; next best step in Mx? → reexamine in one year (reassurance) → periods are typically abnormal in the first year following menarche
- 13F has never had a period + has suprapubic mass + nausea + vomiting; next best step in Mx? → answer = **do beta-hCG** → she's pregnant; this is HY. Correct, girls can get pregnant without ever having had a period → must rule out.
- 14F has massive unilateral breast mass + mom is freaking out bc her sister died of breast cancer → answer = follow-up in six months → **virginal breast hypertrophy** is normal during puberty.
- 15M has unilateral mass behind his nipple +/- tenderness of it → answer = reassurance → physiologic gynecomastia of puberty (higher androgens are aromatized to estrogens).
- Girl is Tanner stage 3; which of the following is true? → **answer = menarche is imminent** → USMLE asks this Q straight up and it's exceedingly HY and frequent.
- 17F + really pad period pain + physical exam is normal → answer = primary dysmenorrhea = prostaglandin hypersecretion (PGF2alpha) → give NSAIDs.
- 23F + really pad period pain + P/E shows nodularity of uterosacral ligaments → answer = endometriosis → do diagnostic laparoscopy.

- 14M is 3rd centile for height + bone age is less than chronologic age; Dx? → constitutional short stature (he'll catch up; his growth curve is just shifted to the right).
- 14M is 3rd centile for height; next best step in Dx? → ask for information about the parents' height trajectory → if they already say in the stem the parents are average height, answer = do bone age.
- 16F is Tanner stage II + wide neck + bone age is equal to chronologic age; Dx? → Turner syndrome → presents with genuine short stature (vignette will often say girl who's 4'11") + Tanner stage I or II + cystic hygroma (lymphatic insufficiency of neck; wide/webbed neck).
- Adult male is 4 feet tall + head and trunk are large in comparison to limbs; Dx? → achondroplasia
- Adult male is 4 feet tall + head and trunk proportional to limbs; Dx? → Laron dwarfism (growth hormone receptor defect → insensitivity to GH).
- Diffusely enlarged uterus + per vaginum bleeding → adenomyosis
- Globular uterus + per vaginum bleeding → leiyomyomata uteri (uterine fibroids)
- 42M has surgery + two days later in hospital he has restlessness + tremulousness + tachycardia + diaphoresis; Dx? → delirium tremens (alcohol withdrawal)
- Tx for DT → long-acting benzo (diazepam or lorazepam or chlordiazepoxide) → USMLE likes chlordiazepoxide.
- 42M has surgery + two days later in hospital he has restlessness + tactile/visual hallucinations; Dx? → alcoholic hallucinosis → on the same spectrum as DT; Tx with long-acting benzo same as DT.
- When is buspirone the answer? → second-line Tx for generalized anxiety disorder (first-line is SSRI).
- MOA of buspirone → serotonin receptor agonist.
- Biochemical disturbance in Addison? → Low Na, high K, low pH, low bicarb
- Biochemical disturbance in Conn syndrome? → High Na, low K, high pH, high bicarb
- Pt has fatigue + normal Na, high K, low pH, low bicarb; Dx? → Addison (sodium can sometimes be normal in aldosterone derangement).
- 22F + BP of 160/110 on multiple office visits + MR angiography of renal vessels confirms diagnosis of fibromuscular dysplasia + labs show normal Na, normal K, normal pH, normal bicarb (Q is: what are her AT-II and aldosterone levels? Answers are up, down, no change for all the different combinations) → **answer = high AT-II + high aldosterone** → learning objective is: it's rare, but biochemistry can be completely normal in aldosterone derangement (Google it if you don't believe me) → **this is on the**

USMLE; if you get a vignette where they 1000% put in your face that a patient has high BP + a confirmed Dx of a cause of hyperaldosteronism, the answer is both AT-II and aldo are high.

- 2-year-old boy has writhing movements in his sleep + periventricular nodules seen on MRI of the head; what else would be seen in this pt? → answer = renal angiomyolipoma or cardiac rhabdomyoma → Dx = tuberous sclerosis (AD).
- Kid + heart tumor = cardiac rhabdomyoma until proven otherwise
- Adult + heart tumor = cardiac myxoma until proven otherwise (ball-in-valve tumor in the left atrium → causes a diastolic rumble that abates when patient is positioned in an unusual way, e.g., on his right side while leaning diagonally).
- 2-year-old boy has cardiac myxoma (correct, not rhabdomyoma) + perioral melanosis (sophisticated way of saying hyperpigmentation around the mouth/lips) + hyperthyroidism; Dx? → answer = **Carney complex** → this is asked on the USMLE → classically triad of cardiac myxoma + perioral melanosis + endocrine hypersecretion (classically bilateral pigmented zona fasciculata hyperplasia resulting Cushing syndrome, but can be hyperthyroidism or growth hormone).
- Biochemical disturbance in DKA → low Na, high serum K (hyperkalemia), low total body K, low bicarb, low pH, low CO2.
- Biochemical disturbance in aspirin toxicity in first 20 minutes: normal O2, low CO2, high pH, normal bicarb → respiratory alkalosis only
- Biochemical disturbance in aspirin toxicity after 20 minutes: normal O2, low CO2, low pH, low bicarb → mixed metabolic acidosis-respiratory alkalosis → one of the 2CK pediatric NBME forms gives a teenage girl who ODed on aspirin 20 minutes ago + they list all of the different acid-base disturbances, and answer is mixed metabolic acidosis-respiratory alkalosis, **not** respiratory alkalosis. So whether you agree with it or not because you think the time frame is too early, I don't know what to tell you, it's the fucking answer on the NBME and everyone gets it wrong, including myself when I answered it. One thing I might point out however is that they said in this Q that the girl had lethargy, which I've noticed having gone thru different NBME Qs repeatedly as a tutor, can non-specifically imply metabolic acidosis → in other words, I've seen various Qs on surg, IM, and peds forms, etc., where there will be, e.g., lactic acidosis, and they'll mention lethargy. I have also seen lethargy in

Addison disease Qs, but I would say **low cortisol causing chronic fatigue syndrome** is the more important association there.

- Patient has fever of 103F + Hb of 7 g/dL + platelet count of 50,000 + neutrophils are few; next best step in Mx? → immediate IV broad-spectrum antibiotics → fever + low neutrophils = febrile neutropenia, aka neutropenic fever; **this is a medical emergency** and means a patient has an infection but no way to fight it off → I have seen plenty of students select "give platelets" as the answer here, no idea why. It's really rare to transfuse platelets, but may be considered with counts under 10-20k if there is spontaneous bleeding. We also tend to transfuse RBCs if under 7 g/dL, but if you get low neutrophils + fever in the same vignette, transfuse RBCs is wrong. I notice the "rule" of transfusing RBCs if Hb is 7 g/dL or lower causes students to get Qs wrong; think of it as a general propensity for transfusion, rather than as a mandatory answer for the Q.
- AV-nicking on fundoscopy → answer = hypertensive retinopathy
- Tx for hypertension? → if patient has pre-diabetes, diabetes, or any cardiovascular/cerebrovascular disease of any kind → **answer = ACEi or ARB first.** These agents decrease morbidity and mortality in these patient groups. If patient has none of the above (i.e., your typical fat American middle-age male who's a little overweight but otherwise just has essential hypertension), the answer = **HCTZ or dihydropyridine CCB.** You might think that's really weird (i.e., "why not just give an ACEi or ARB anyway to anyone if they're good for morbidity/mortality?"), but the basis is: you're not going to live to 120 just because you start taking a statin when it's not indicated; well the same is true here: there's no evidence of further improvement or morbidity/mortality in pts without the above risk factors if started on ACEi or ARB). This knowledge about how to Tx HTN is HY for FM shelves in particular.
- 32F + pedal + forearm edema after commencing anti-hypertensive agent; Dx? → answer = fluid retention / edema caused by dihydropyridine CCB (e.g., nifedipine) → really HY side-effect of d-CCBs!
- Side-effects of thiazides → hyperGLUC → hyperglycemia, -lipidemia, -uricemia, calcemia.
- Whom should you never give thiazides to? → prediabetics or diabetics → will push people into type II DM and make current DMs worse. One of the worst/frequent pharmacologic mistreatments. Also don't give to pts with Hx of gout (bc of hyperuricemia risk).
- Diabetic pt on HCTZ for HTN → take them the fuck off the thiazide and put them on an ACEi or ARB.

- Important use of thiazide apart from HTN management in select patients → decreased risk of nephro- / ureterolithiasis (stones) because they cause hypocalciuria (and hence hypercalcemia).
- Delirium in patient on thiazide → think hypercalcemia → high Ca is HY USMLE cause of CNS dysfunction → "hypercalcemic crisis."
- Renal issue + hypercalcemia; Dx? → nephrogenic diabetes insipidus → on the NBME.
- CNS disturbance due to an electrolyte problem → high or low sodium; high calcium
- Cardiac disturbance (arrhythmia) due to an electrolyte problem → high or low potassium or calcium
- Low calcium or potassium not responding to supplementation → check serum magnesium (low Mg can cause low Ca and K refractory to supplementation
- Who gets low Mg → alcoholics (dietary deficiency; nothing malabsorptive or magic; they just drink too much and fill up on EtOH bc it's 7kcal/gram).
- Nutrition calorie counts: carbs + protein = 4kcal/g; fat = 9 kcal/g; EtOH = 7kcal/g.
- Tx for pulmonary hypertension → most patients respond to dihydropyridine CCBs (e.g., nifedipine).
- If patient fails the dCCB test, can try agents like bosentan or sildenafil.
- MOA of bosentan → endothelin-1 receptor antagonist
- 28F non-smoker has loud P2 + RVH; Dx? → primary pulmonary hypertension
- Which of the following is true in the above 28F? → answer = "increased vascular expression of endothelin 1." → if you know bosentan can treat, then inferring this is easy.
- VSD is repaired with a prosthetic patch; how will LV, RV, and LA pressures change? → answer = LV pressure goes up, RV pressure goes down, LA pressure goes down. → student says "the LA one is weird tho why is that?" Because if you decrease the LV → RV shunt, then there's less blood circulating back to the LA → decreased LA preload → decreased LA pressure.
- Kid is given over-the-counter med by his mom for a cold + gets mental status changes; Dx? → anticholinergic delirium caused by first-generation antihistamine (e.g., diphenhydramine, chlorpheniramine).
- 22M takes a drug + gets nystagmus + bellicosity (wants to fight) → **answer = PCP.**
- 22M takes a drug + gets mutism + has constricted pupils → answer = PCP. Fucking weird but it's on the psych NBME for 2CK. If you don't believe me, you can Google "pcp mutism constricted pupils."

- 2-year-old boy running + playing with 8-year-old sister + they were holding hands and he fell + now he holds arm pronated by his side; Dx? → **nursemaid's elbow** → radial head subluxation
- Tx for nursemaid's elbow → hyperpronation OR gentle supination (both are correct answers; only one will be listed).
- Kid falls on outstretched arm + pain over anatomical snuffbox; Dx? → scaphoid fracture
- Kid falls on outstretched arm + pain over anatomical snuffbox; next best step in Mx? → x-ray
- Kid falls on outstretched arm + pain over anatomical snuffbox + x-ray is negative; next best step in Mx? → **thumb-spica cast** → x-ray is often negative in scaphoid fracture; must cast to prevent scaphoid avascular necrosis → re-x-ray in 2-3 weeks.
- When is figure-of-8-strap the answer? → clavicular fracture
- What part of the clavicle gets fractured easiest? → middle-third
- First Tx for carpal tunnel syndrome in patient who can't stop offending activity (e.g., office worker) → **wrist splint first;** then **triamcinolone (steroid) injection into the carpal tunnel**; do not select anything surgical as it's always wrong on the USMLE; NSAIDs are a wrong answer and not proven to help
- Tx for cubital tunnel syndrome → elbow splint
- What is cubital tunnel syndrome → ulnar nerve entrapment at elbow → presents similarly to carpal tunnel syndrome but just in an ulnar distribution and involves the forearm.
- What is Guyon canal syndrome → ulnar nerve entrapment at the wrist → **hook of hamate fracture** or chronic handle bar impaction in avid cyclists.
- Most likely organism causing impetigo → **S. aureus now exceeds Group A Strep** for non-bullous ("regular") impetigo; for bullous, S. aureus has always exceeded S. pyogenes.
- Tx of impetigo → topical mupirocin
- Golden crusty lesions around the mouth in school-age child → impetigo, not HSV.
- 32F + sharply demarcated fiery red lesion extending from the knee to ankle + fever of 101F → answer = **erysipelas** → a Dx students never remember well → not as bad as cellulitis → erysipelas is infection of superficial dermis +/- dermal lymphatics, whereas cellulitis is hypodermis; the superficial nature of erysipelas gives it a well-demarcated, fiery appearance, whereas cellulitis is more diffuse and pink.
- Most common organism for erysipelas? → Group A Strep far exceeds S. aureus (but do not neglect the latter).

- Most common organism for cellulitis? → S. aureus exceeds Group A Strep.
- Treatment for erysipelas + cellulitis → oral dicloxacillin or oral cephalexin → both agents cover Staph + Strep; penicillin only covers Strep.
- Severe skin infection involving fascial planes + cutaneous crepitus; organism? → Clostridrium perfringens causing necrotizing fasciitis (polymicrobial, but the C. perfringens causes the gas gangrene leading to cutaneous emphysema / crepitus).
- Tx for nec fasc → surgical debridement + IV broad-specrum Abx with anaerobic coverage.
- Perineal gangrene in 50M diabetic → Fournier gangrene → do surgical debridement.
- 17M comes to emergency with cellulitis + getting worse + holding amoxicillin canister he got from GP; Dx? → improper Abx treatment; should have received oral dicloxacillin or oral cephalexin outpatient → cellulitis must have been caused by Staph not Strep.
- Above 17M; what do you do? → Stat dose of IV flucloxacillin or IV cephazolin (inpatient equivalents of oral dicloxacillin + oral cephalexin).
- Why doesn't amoxicillin or penicillin cover Staph? → Most community Staph (not MRSA; just MSSA) produces beta-lactamase, so much give beta-lactamase-resistant beta-lactam (diclox and fluclox are steric; drugs like nafcillin and oxacillin are typically used for osteomyelitis; 6 weeks nafcillin is classic for confirmed MSSA endocarditis).
- Renal issue + beta-lactam or cephalosporin → interstitial nephropathy (aka tubulointerstitial nephritis) → WBCs on dipstick (eosinophils on Homer-Wright staining). Stress incontinence → weakened pelvic floor muscles resulting in loss of urine with increased abdominal pressure (coughing, sneezing, laughing) → Hx of multiple pregnancies classic, but often too easy of a descriptor and they won't say that → they'll say there's "downward movement of the vesicourethral junction with coughing"; next best step in Mx? → pelvic floor (Kegel) exercises → if insufficient, do mid-urethral sling; **do not** give medications for stress incontinence (HY!).
- USMLE might ask you which muscle is **not** strengthened by Kegel exercises → student then proceeds to have two thoughts: 1) "wtf, I'm supposed to know Kegel exercises at that high level of detail?" and 2) couldn't *any* muscle not be strengthened by Kegel exercises; I mean, the deltoid wouldn't be for instance." → **answer to this Q = internal anal sphincter** → even if you have zero clue about Kegel exercises, bear in mind **internal sphincters (urethral + anal) are under sympathetic control → you**

can't voluntarily strengthen a muscle not under somatic (voluntary) control; in case you're curious though, Kegel strengthens levator ani (which comprises pubococcygeus, puborectalis, and iliococcygeus).

- Urge incontinence → answer = "hyperactive detrusor," or "detrusor instability" → needs to run to the bathroom when sticking a key in the front door; needs to run to bathroom when opening car door; answer in multiple sclerosis + menopause (part of vasomotor Sx); can be idiopathic; answer = give oxybutynin (anti-muscarinic) or mirabegron (beta-3 agonist); **once again, do not give these drugs in stress incontinence.**
- Overflow incontinence → answer in diabetes and BPH → neurogenic bladder caused by myelin damage from sorbitol (glucose enters myelin, causing osmotic damage); in BPH, merely due to outlet obstruction → leads to detrusor burnout; **in overflow incontinence, postvoid volume is high (i.e., 300-400 mL in USMLE Qs);** normal should be <50-75 mL; for diabetic bladder, answer = bethanechol (muscarinic agonist); for BPH, **insert catheter first always.**

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HY USMLE REVIEW
PART III